Innovation partner

H2S

Medla partner

YOURSTORY

AI for Bharat Hackathon

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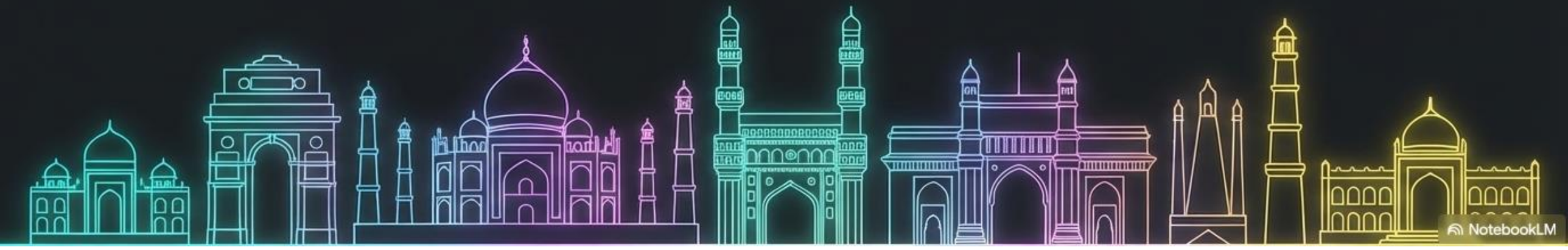

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AI for Bharat Hackathon

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Team Name: Choco Warriors

Team Leader: Shashank Hara Sridhar

Problem Statement: Design an AI solution that improves efficiency, understanding, or support within healthcare or life-sciences ecosystems.

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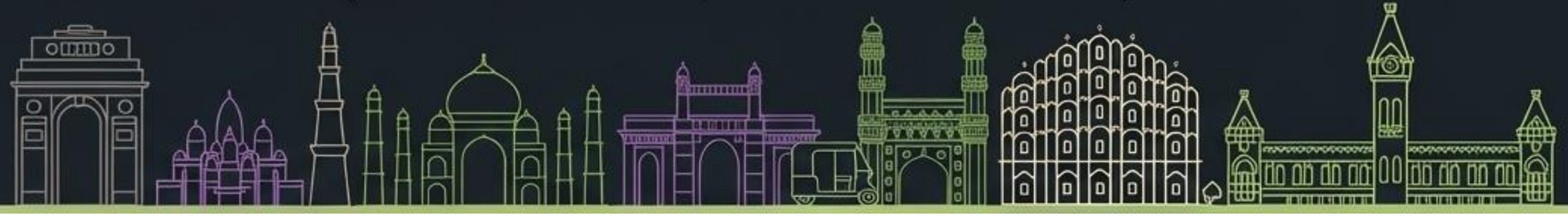
Brief about the Idea:
The Hospital Review Platform

A web application designed to improve healthcare transparency for income-constrained individuals. The platform aggregates verified patient reviews backed by authentic documents (payment receipts, insurance claims, discharge summaries) to help users find the best hospitals based on their specific needs, budget, and insurance coverage.

Transparent Facts Exposed:

- Doctor Qualifications: Verified specialties and patient outcomes.
- True Surgery Costs: Actual out-of-pocket costs paid by real patients.
- Insurance Coverage: Exact amounts insurance companies actually paid for claims.
- Treatment Timelines: Expected duration for procedures and recovery.



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Why AI is required?

Overcoming Traditional Limitations

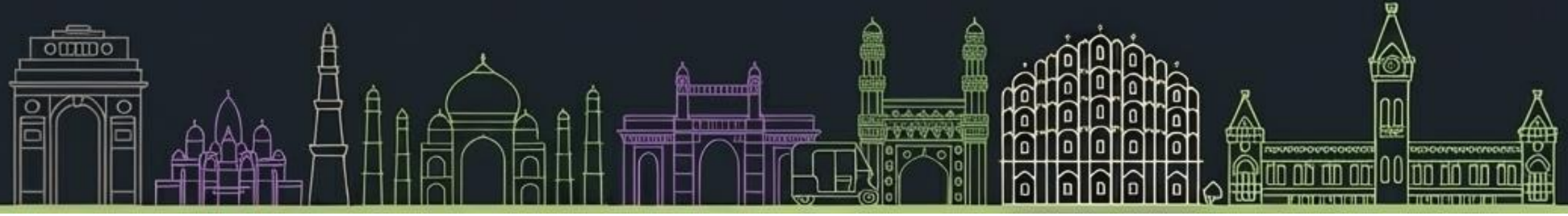
Traditional keyword search fails completely when faced with unstructured medical context and specific patient anxieties.

Natural Language Translation

Generative AI translates complex medical jargon into understandable terms and accurately interprets natural language needs (e.g., matching a search for 'affordable cardiac care' with the right specialists).

Semantic Understanding

AI decomposes complex, multi-variable queries and semantically searches over 10,000+ unstructured patient reviews to deliver purely experience-based, highly relevant hospital recommendations.



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How AWS services are used?

Multi-Agent Intelligence

Powered by Amazon Bedrock for complex query breakdown, contextual routing, and synthesizing insights.

Vector Search & Embeddings

Amazon OpenSearch Service leveraged for hyper-fast semantic querying of unstructured patient reviews.

Automated Document Extraction

Amazon Textract for OCR on medical bills, and Comprehend Medical for identifying specialized medical entities.

Serverless Orchestration

AWS Step Functions flawlessly orchestrating the document validation and ingestion pipeline.

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Scalable Healthcare Architecture on AWS

AWS Amplify Deployment

Streamlines the continuous deployment and hosting of the patient-facing application.

AWS S3 for Patient Verification

Securely stores uploaded patient PDFs required for identity and insurance verification.

AWS DynamoDB for Document Storage

Houses unstructured JSON documents for flexible and scalable patient record management.

AWS AppSync Real-Time Streaming

Enables low-latency, real-time data streaming between patients and medical agents.

AWS Location Services

Calculates precise distances between a user's current location and the nearest hospital.

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What value the AI layer adds?

100% Verified Proof

Delivers absolute financial transparency by tying every review to an AI-validated receipt or insurance claim, eliminating fake reviews entirely.

540x Process Acceleration

Automated, 20-second document validation drastically reduces manual review times, allowing instant publication of trusted patient experiences.

Real-Time Transparency

Live agent activity streams via AWS AppSync build profound user trust by showing exactly how the AI arrives at its hospital recommendations.

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List of features offered

Amazon Bedrock

Natural Language Search

Ask complex medical questions like a consultant.

AWS Verified Access

Verified Reviews

Marked 'Verified' only after AI validates proof of visit.

AWS Cost Explorer

Cost Estimator

Real-time visibility into exact amounts paid by other patients.

AWS HealthLake

Insurance Compatibility

Filter hospitals by successful claim histories.

Amazon Comprehend Medical

Smart Summaries

AI-generated pros and cons extracted from thousands of patient profiles.

Amazon Rekognition

Fraud Detection

Automated flagging of manipulated images and mismatched medical documents.

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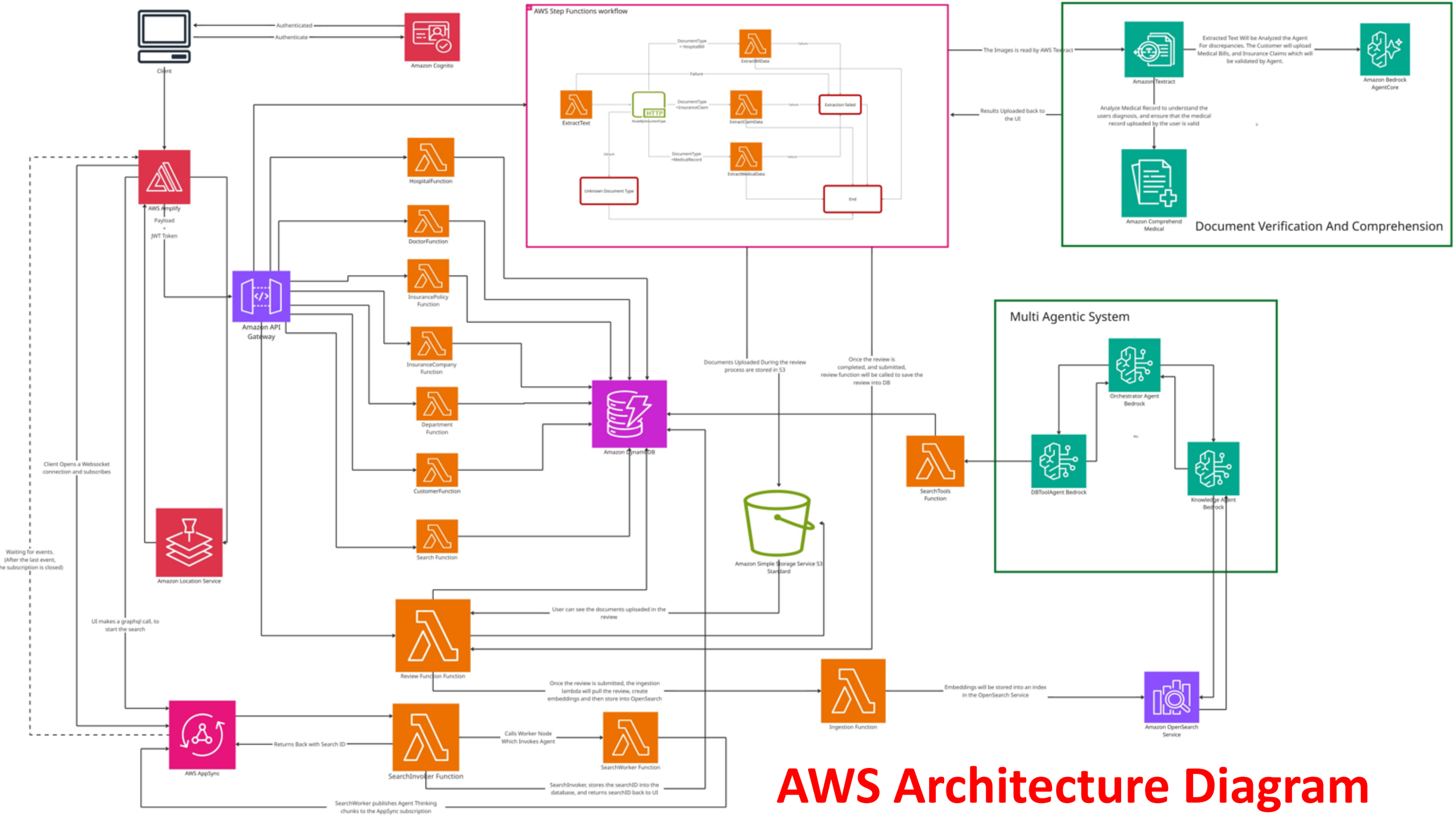
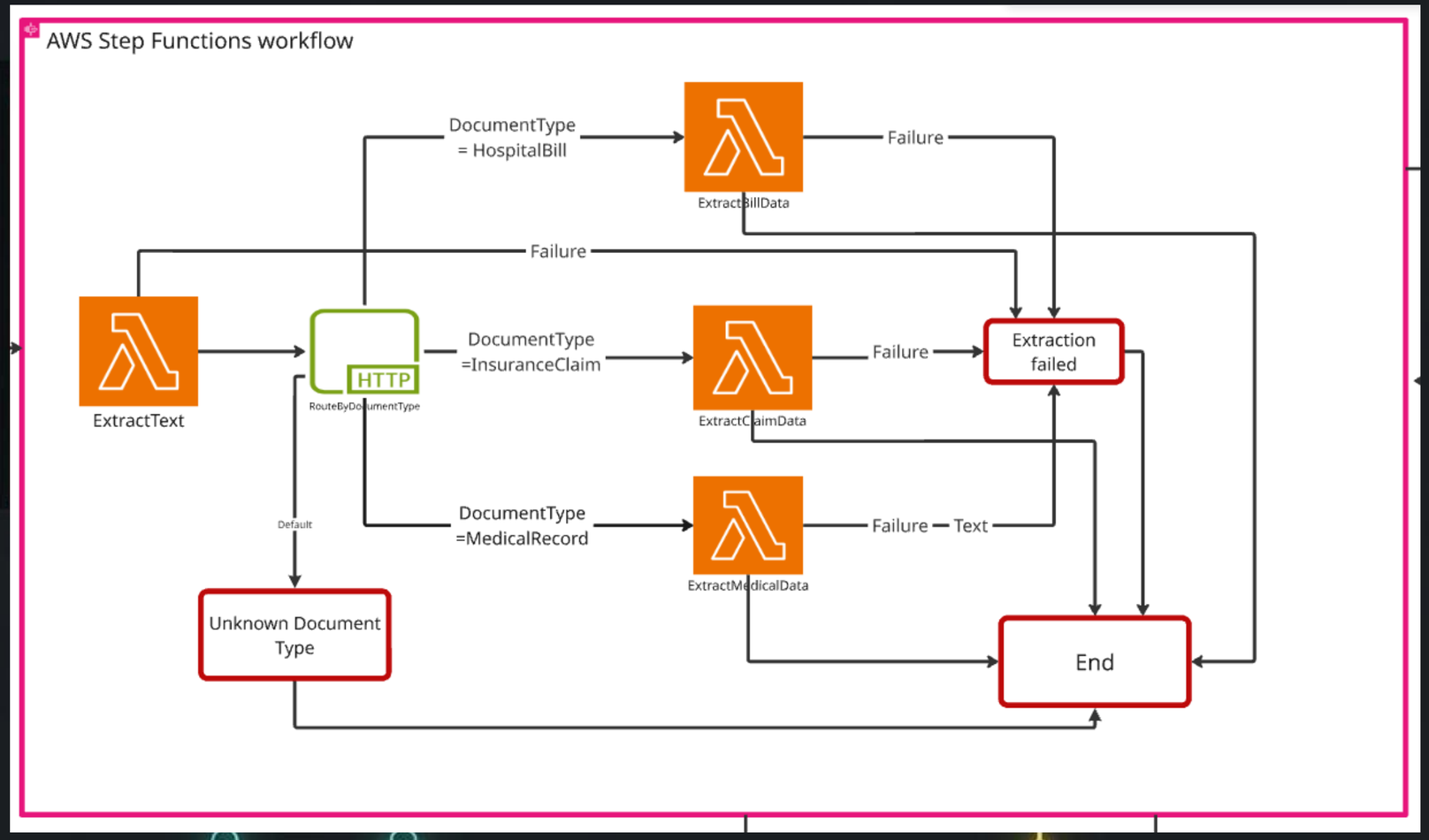


AWS Architecture Diagram

Zoom 1: Automated Document Verification

- **Trigger:** Patient uploads hit Amazon S3, triggering the AWS Step Functions workflow.
- **Routing:** The state machine routes documents by type (Hospital Bill vs. Medical Record).
- **Extraction:** Amazon Textract extracts raw OCR data and key-value pairs.
- **Validation:** Amazon Bedrock and Comprehend Medical validate the entities, identify discrepancies, and discrepancies, and finalize extraction.



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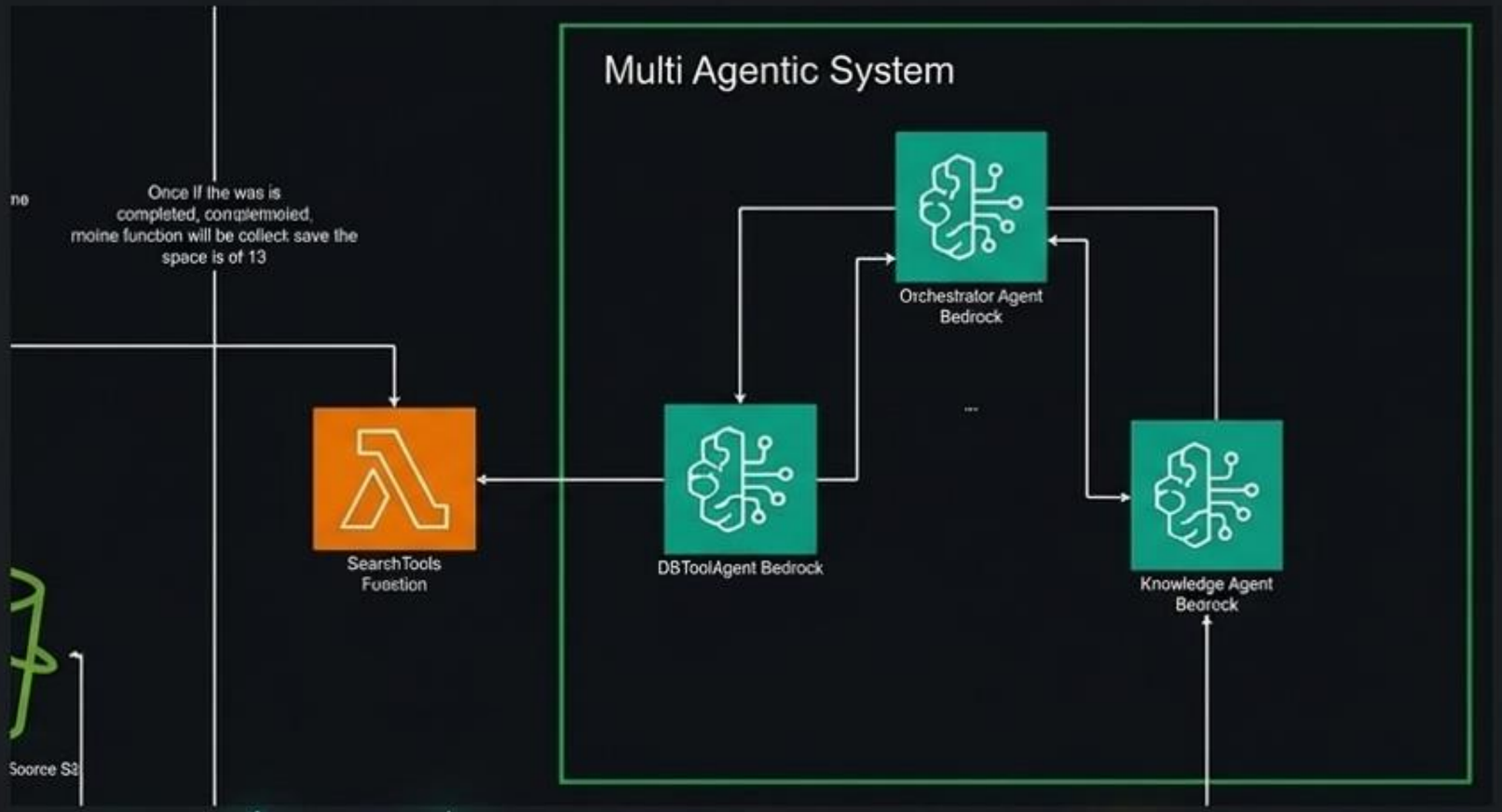
Zoom 2: The Multi-Agentic Brain

- **Orchestrator Agent:** The core router that breaks down complex natural language queries (e.g., affordable pediatric care with Blue Cross).
- **DB Tool Agent:** Executes specific functions to fetch structured data (cost metrics, insurance matrices) from DynamoDB.
- **Knowledge Agent:** Runs semantic analysis over the unstructured review database to find contextually relevant patient experiences.



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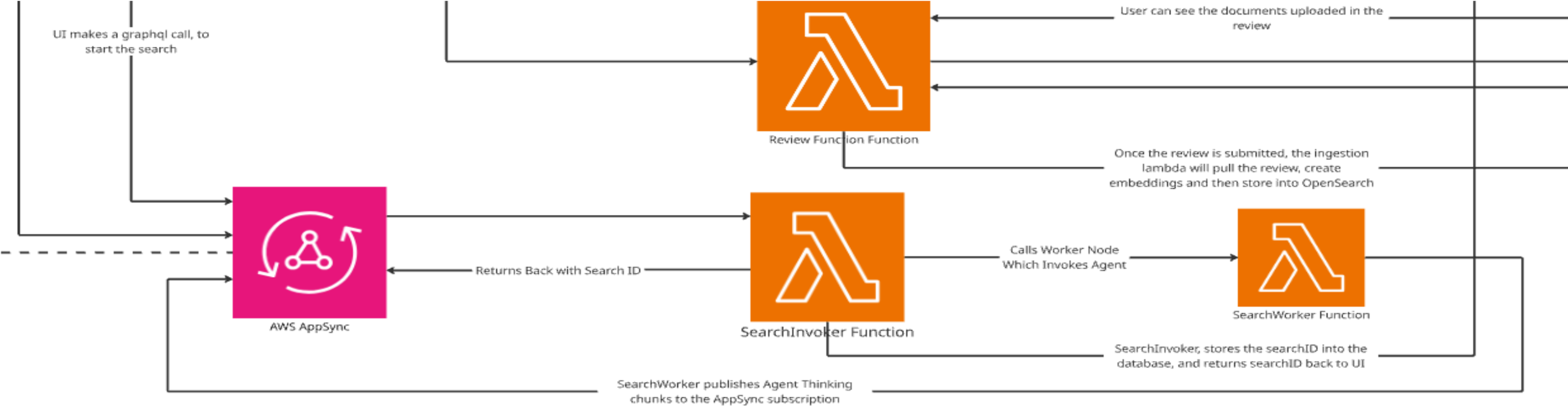

Zoom 3: Semantic Search & Real-Time Sync

- **Ingestion:** Verified reviews are passed to a Lambda that generates vector embeddings (Titan Embed) and stores them in OpenSearch Serverless.
- **Retrieval:** The SearchWorker Lambda coordinates with the multi-agent system to query this vector index.
- **Live Streaming:** AWS AppSync establishes WebSockets, streaming live agent activity and personalized search results directly back to the client UI.

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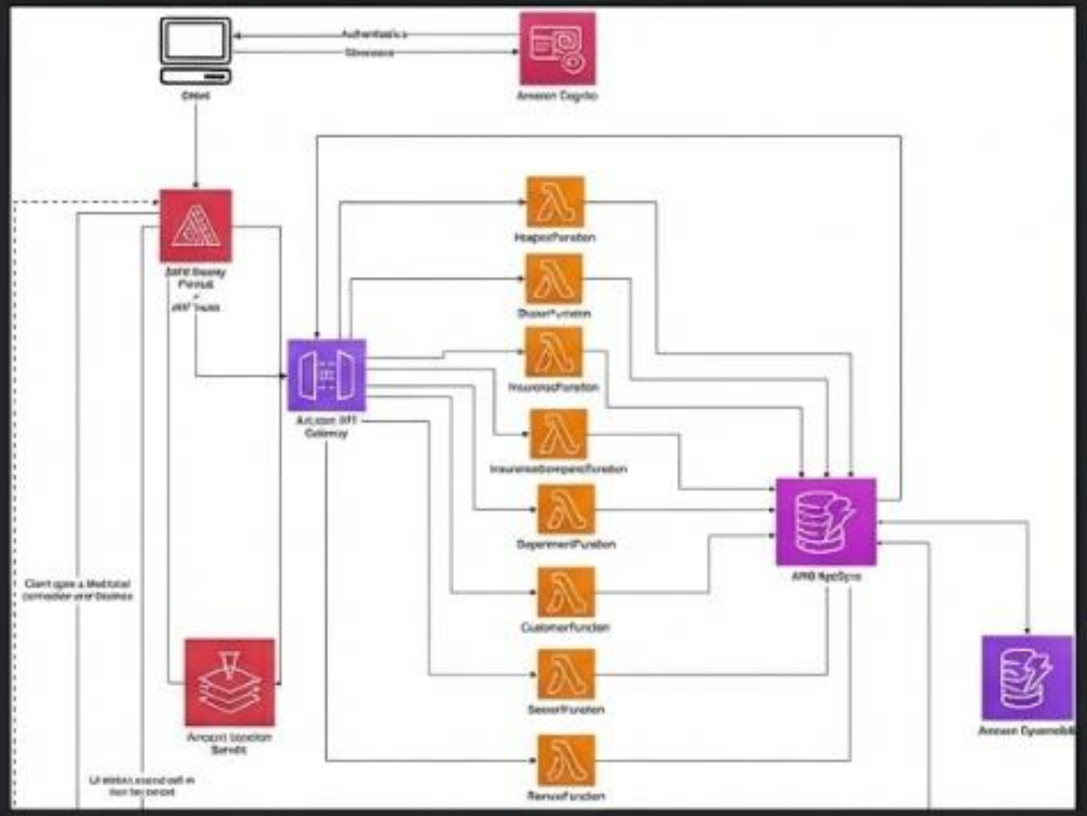
Architecture diagram of the proposed solution: Part 1



Core App & Data Routing:

- **Client & Authentication:** Secure user authentication managed by Amazon Cognito.
- **API & Routing:** Amazon API Gateway routes RESTful requests to dedicated serverless Lambda functions handling CRUD operations for Hospitals, Doctors, Insurance, and Reviews.
- **Real-Time Streaming:** AWS AppSync establishes WebSocket canet connections to stream live AI agent search activity directly back to the client interface.
- **Core Storage:** Amazon DynamoDB acts as the primary NoSQL data store.
- **Mapping:** Amazon Location Service provides interactive spatial data.

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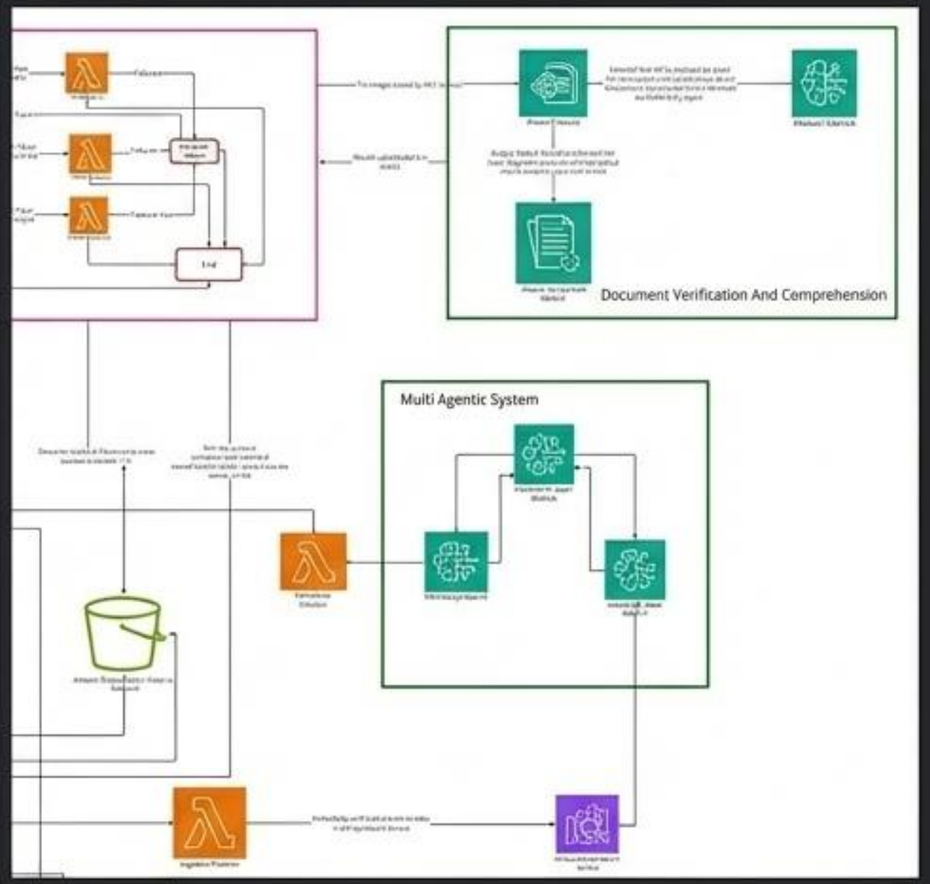
Architecture diagram of the proposed solution: Part 2

AI Multi-Agent & Verification:

- **Document Verification Pipeline:** AWS Step Functions orchestrates a rigorous state machine where Amazon Textract pulls OCR data and Comprehend Medical verifies entities.
- **Multi-Agentic System (Amazon Bedrock):**
  Multi-Agentic System (Amazon Bedrock):
  - **Orchestrator Agent**: Breaks down complex patient queries.
  - **DB Tool Agent**: Executes functions to fetch structured cost/insurance data.
  - **Knowledge Agent**: Queries the OpenSearch index for experiential data.
- **Semantic Search Engine**: An Ingestion Lambda creates vector embeddings of verified reviews, storing them in Amazon OpenSearch Service for relevance-based querying.



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Process flow diagram

From Raw Upload to Verified Insight

1. Patient Upload

User uploads a medical bill or insurance claim to Amazon S3.

2. Workflow Triggered

AWS Step Functions pipelines orchestrates the validation pipeline.

3. Data Extraction

Extraction of entities, costs, and medical data.

4. AI Validation

Cross-referencing extracted data for discrepancies and authenticity.

5. Verified Publication

Approved document is indexed, and review is marked 100% verified.

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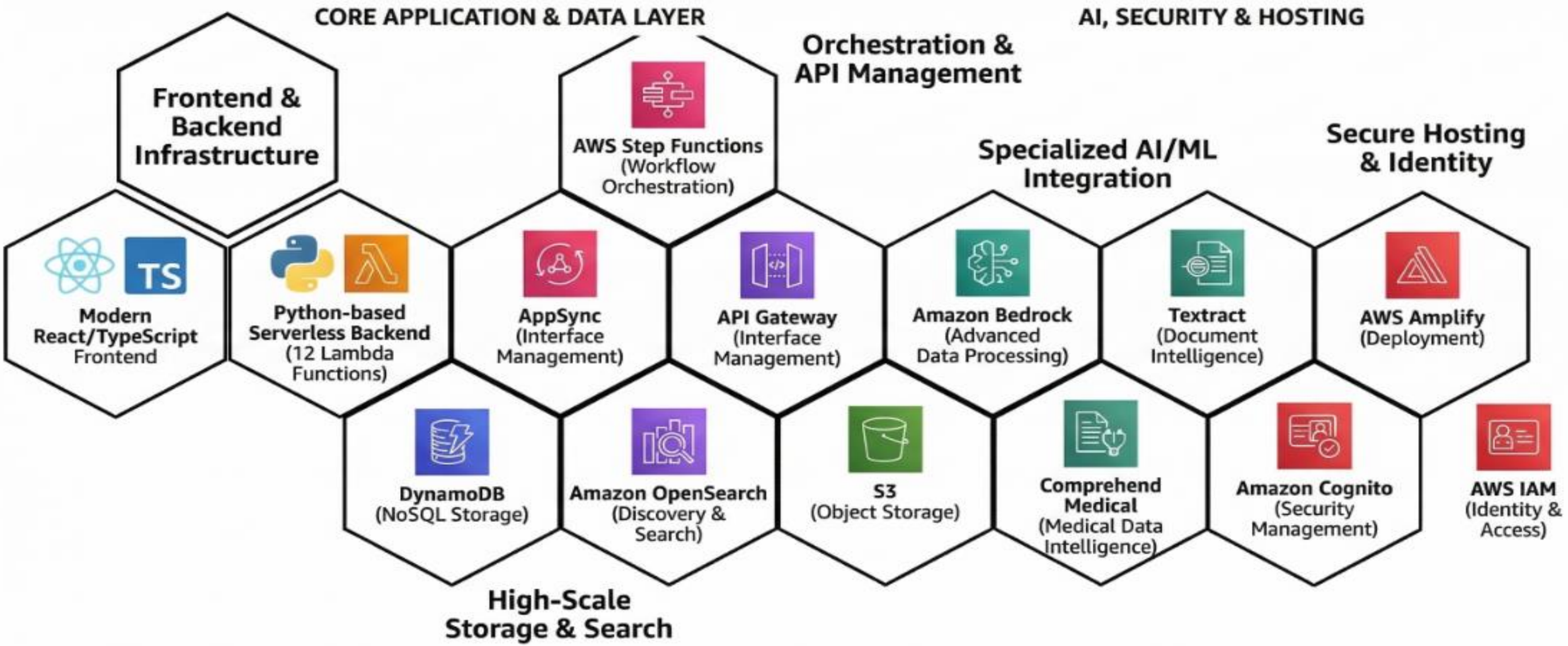
Technologies utilized in the solution:

Modern Cloud-Native Solution Architecture



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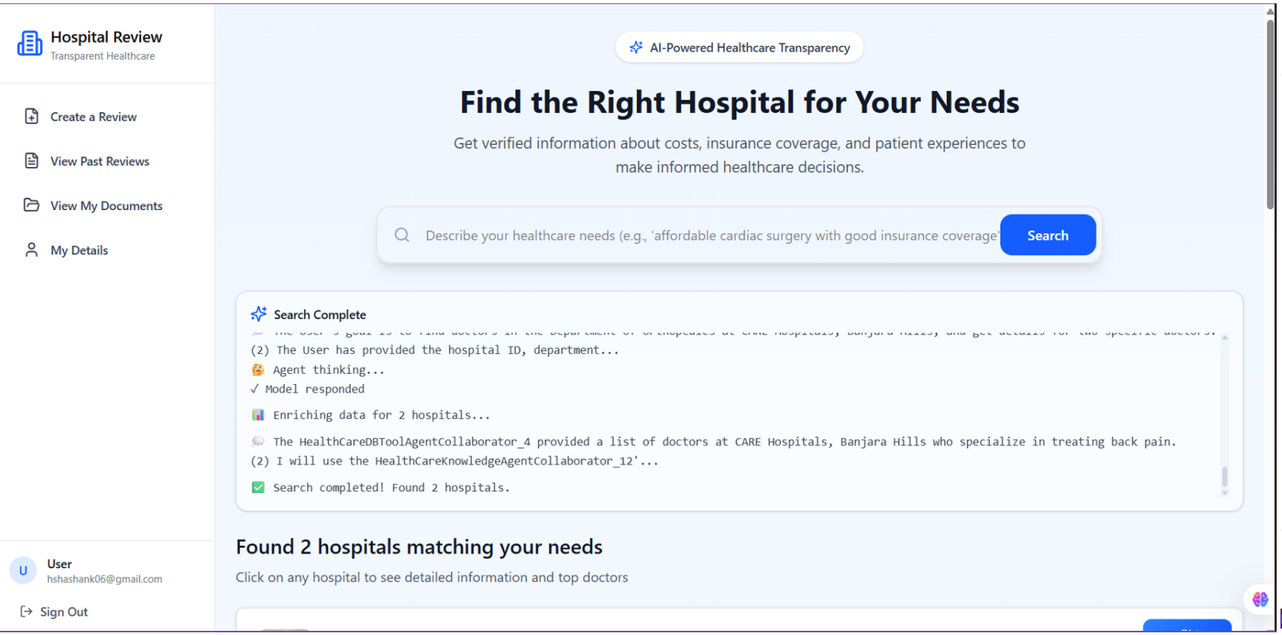
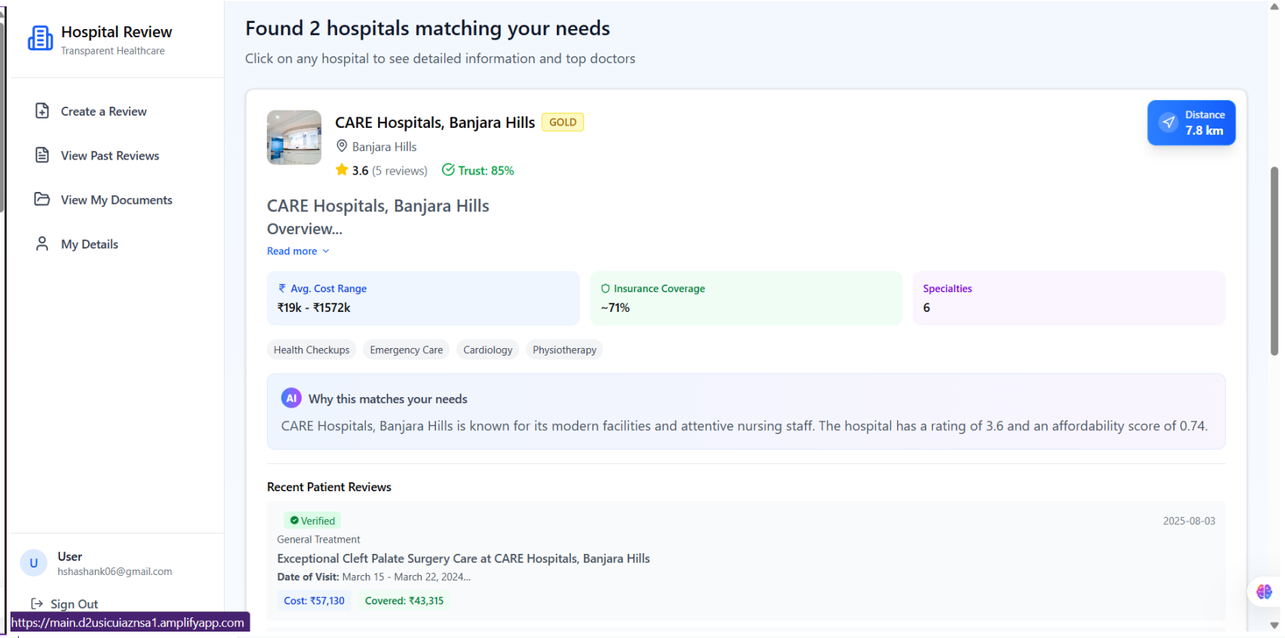
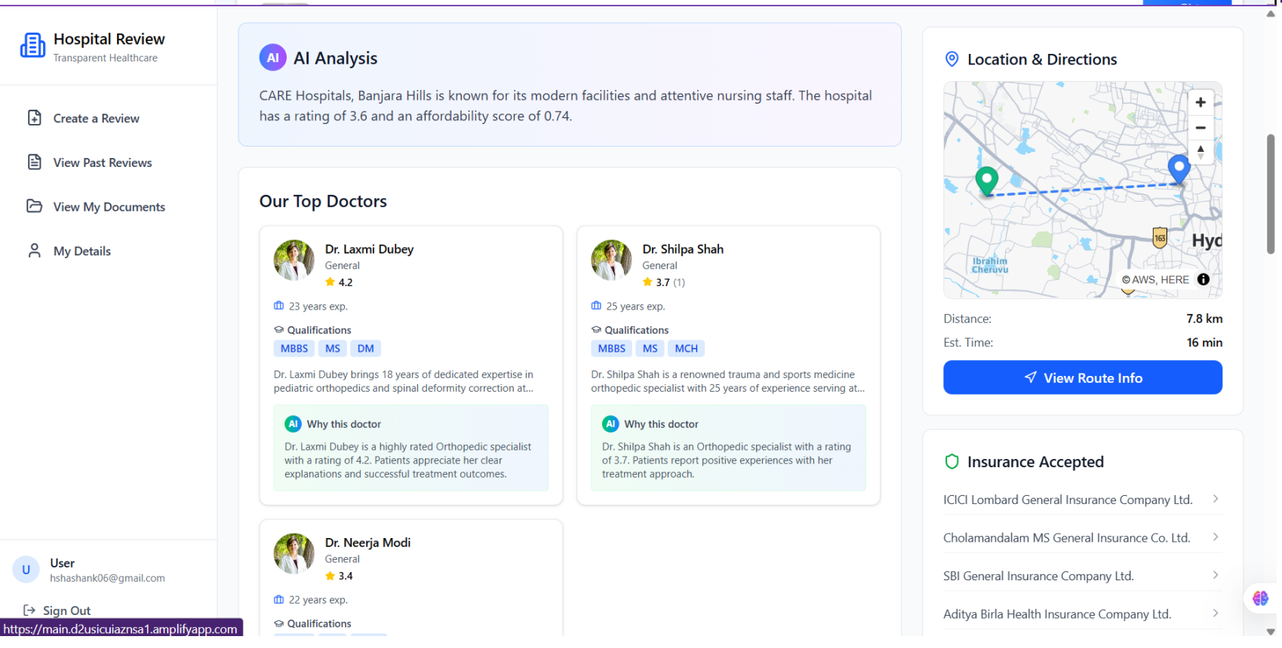
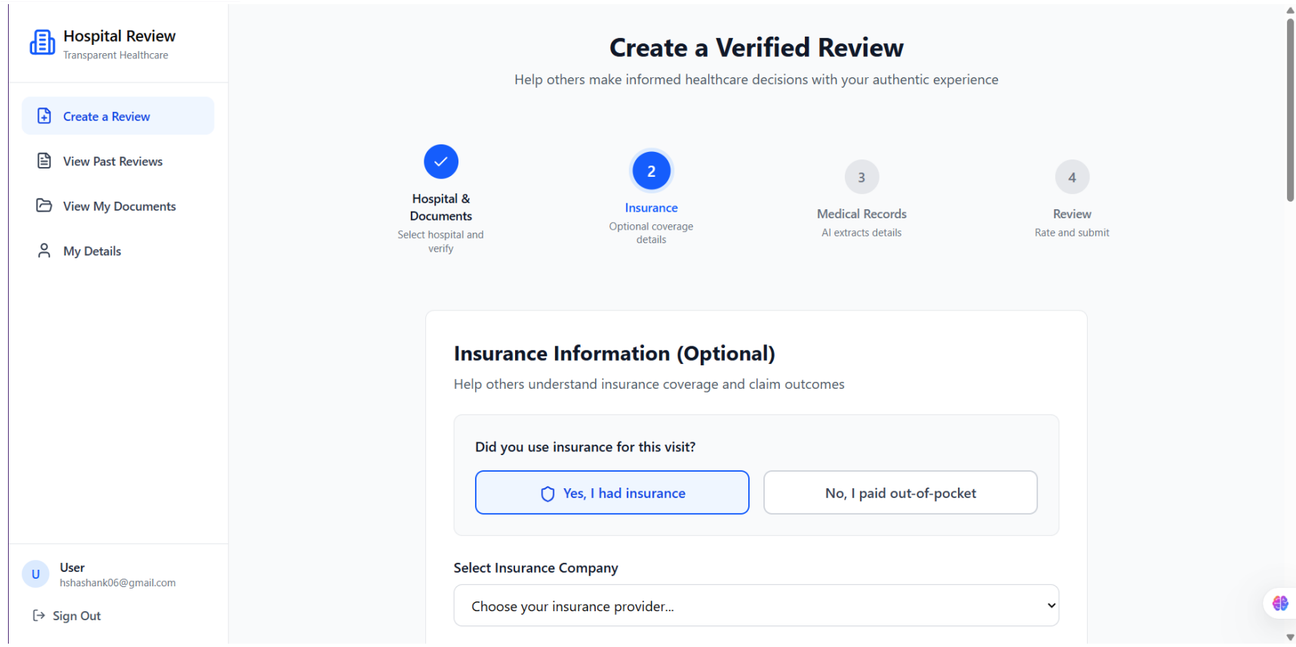
SNAPSHOTS OF THE PROTOTYPE

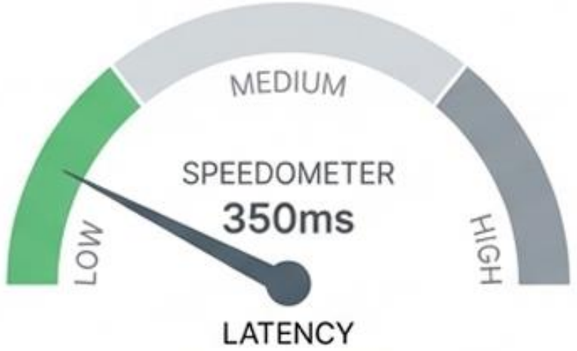








Prototype Performance report/Benchmarking:



Speed & Latency

Initial response in just 350ms. Complete, multi-agent AI semantic search completes in 18-25 seconds (fully streamed in real-time to the user).

AI Quality & Accuracy

95% Search Relevance (compared to ~60% on standard map applications), with excellent context understanding and cost personalization.

Scale & Reliability

Architecture tested to handle up to 10,000 concurrent users using Lambda provisioned concurrency and DynamoDB auto-scaling.

Uptime

99.9% uptime over 30 days with < 0.5% error rate across all AWS services.

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Additional Details/Future Development

Privacy-Preserving Evidence

Display grounded documents as public evidence with advanced, automated blurring of all Personally Identifiable Information (PII).

National Identity Integration

Implement direct Aadhar and Ayushman Bharat authentication for seamless patient onboarding and subsidy tracking.

Agency-Level Verification

Establish direct API integrations to verify medical records and insurance claims against central government and insurance provider agencies in real-time.

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Prototype Assets:

SCAN ME

GitHub Public Repository Link
https://github.com/hshashank06/aws-hackathon-2026

Scan Me

Demo Video Link (Max: 3 Minutes)
https://www.youtube.com/watch?v=5clatEqXEtE

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Thank You

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